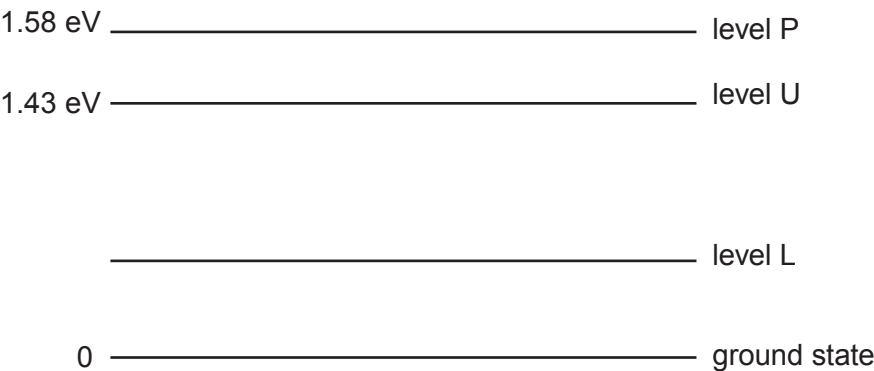


8. The following diagram shows the energy levels for the amplifying medium in a 4-level gas laser.



(a) The laser transition between level U and level L produces stimulated photons of wavelength 1.06×10^{-6} m. Determine the value of energy level L in **eV**. [3]

.....

.....

.....

.....

.....

(b) When explaining the operation of this laser, a student writes the following:

Pumping is required to move electrons from ground level to level P which is a metastable state. This creates a population inversion between P and ground which is needed for the laser to work.

State **two** mistakes in the student's explanation. [2]

.....

.....

.....



Examiner
only

- (c) Semiconductor lasers are regularly used in DVD players and bar code scanners. State **two** advantages that these semiconductor lasers have over gas lasers. [2]

.....

.....

.....

- (d) As electronic technology advances, outdated electronic devices and their components are discarded. It has been estimated that 50 million tonnes of electronic waste is produced each year and roughly 80% of this is finding its way into landfill. Discuss the effects this electronic waste may have on society. [2]

.....

.....

.....

.....

9

END OF PAPER

